



AI-Powered Behavioral Anomaly Detection Report

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Executive Summary

Metric	Value
Total Access Logs Analyzed	4878
Total Anomalies Detected	278
Critical Alerts	0
High Severity Alerts	0
Average Risk Score	95.8%
Report Period	Last 30 Days

Threat Summary

The system analyzed 4878 access events and detected 278 anomalies. No critical threats were identified. Average risk score: 95.8%.

Detected Attack Types

Attack Type	Count	Percentage
Brute Force	70	25.2%
Insider Drift	62	22.3%
Low And Slow Exfiltration	42	15.1%
Lateral Movement	39	14.0%
Credential Stuffing	30	10.8%
Impossible Travel	18	6.5%
Device Spoofing	15	5.4%
Unknown	2	0.7%

Risk Score Distribution

Risk Level	Count
Low (0-25)	0
Medium (25-50)	2
High (50-75)	8
Critical (75-100)	268

Model Performance

Metric	Value
Model Type	IsolationForest
Accuracy	99.9%
Precision	99.3%
Recall	99.3%
F1 Score	99.3%
Training Samples	4869

Recommendations

- Investigate Critical Alerts Immediately** — All alerts with risk scores above 75 should be triaged by the SOC team within 1 hour.
- Enable Multi-Factor Authentication** — Enforce MFA for all user accounts to mitigate credential stuffing and brute force attacks.
- Implement Geo-IP Restrictions** — Block or challenge logins from countries not in the organization's operational geography.
- Deploy Device Trust Framework** — Register and monitor device fingerprints to detect spoofing attempts.
- Review Access Policies** — Audit user access to sensitive resources and enforce least-privilege principles.
- Enhance Monitoring for Off-Hours Activity** — Increase alerting sensitivity for access events outside normal business hours.
- Retrain Model Periodically** — Retrain the anomaly detection model monthly to adapt to evolving behavioral baselines.

8. Conduct Incident Response Drills — Regular tabletop exercises ensure the team can respond effectively to detected threats.

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